NL Refinitiv process

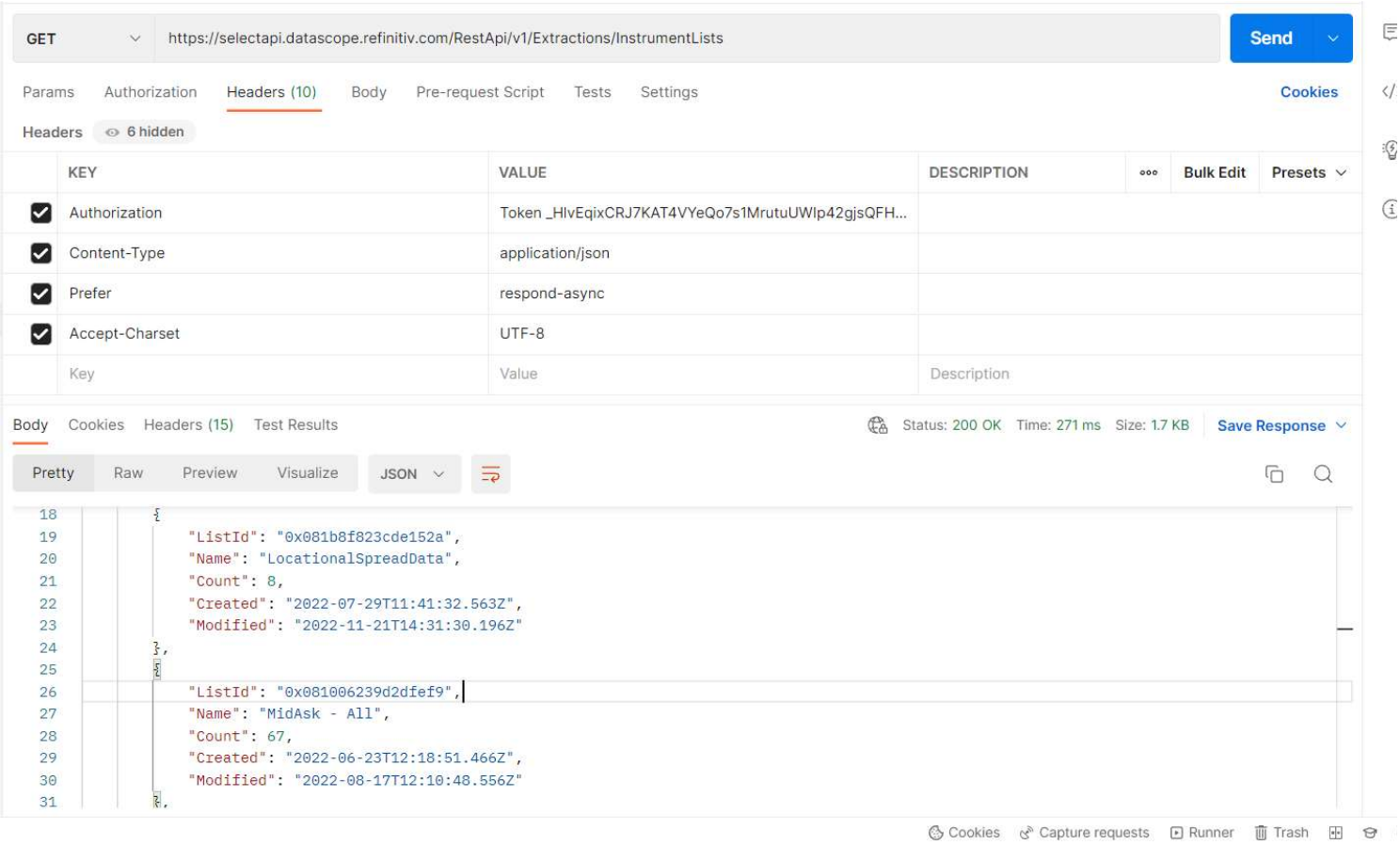
Last updated by | Vlad Suciu | Feb 6, 2023 at 3:09 PM GMT+2

The purpose of this process is to load the market prices from Refinitiv and store them in the ExternalParties database.

There are imported two types of data from different tamplates: MidAsk data and LocationalSpread data. These processes run every working day, LocationalSpread at 8:00 AM UTC time and MidAsk at 8:10 AM UTC time.

The data is taken from the source for the last 7 days and stored in the database for 2 years.

For MidAsk we import 67 files and for LocationalSpread 8 files.



List of identifiers for MidAsk:

refinitiv-raw-data > MidAsk - All

Name	Access Tier	Access Tier Last Modified	Last Modified	Blob Type	Content Type	Size	Status	Deletion ID	Remaining Days	Deleted Time	Le
EDXEBEBMc1			10/14/2022 7:21 AM		Folder		Active				
EDXEBEBMc2			10/14/2022 7:31 AM		Folder		Active				
EDXEBEBMc3			10/14/2022 7:34 AM		Folder		Active				
EDXEBEBMc4			10/14/2022 7:21 AM		Folder		Active				
EDXEBEBQc1			10/14/2022 7:21 AM		Folder		Active				
EDXEBEBQc2			10/14/2022 7:21 AM		Folder		Active				
EDXEBEBQc3			10/14/2022 7:21 AM		Folder		Active				
EDXEBEBQc4			10/14/2022 7:35 AM		Folder		Active				
EDXEBEBYc1			10/14/2022 7:35 AM		Folder		Active				
EDXEBEBYc2			10/14/2022 7:35 AM		Folder		Active				
EDXEBEBYc3			10/14/2022 7:23 AM		Folder		Active				
EDXENLBMc1			10/14/2022 7:16 AM		Folder		Active				
EDXENLBMc2			10/14/2022 7:21 AM		Folder		Active				
EDXENLBMc3			10/14/2022 7:30 AM		Folder		Active				

Showing 1 to 67 of 67 cached items

List of identifiers for LocationalSpread:

refinitiv-raw-data > LocationalSpreadData

Name	Access Tier	Access Tier Last Modified	Last Modified	Blob Type	Content Type	Size	Status	Deletion ID	Remaining Days	Deleted Time	Lease S
PEGDA			10/11/2022 5:24 PM		Folder		Active				
PPEGMc1			10/11/2022 5:24 PM		Folder		Active				
PTTFMc1			10/11/2022 5:24 PM		Folder		Active				
TFMBMc1			10/11/2022 5:24 PM		Folder		Active				
TTFDA			10/11/2022 5:24 PM		Folder		Active				
ZTPDA			10/11/2022 5:24 PM		Folder		Active				
ZTPLDA			10/11/2022 5:23 PM		Folder		Active				
ZTPMc1			10/11/2022 5:24 PM		Folder		Active				

Showing 1 to 8 of 8 cached items

For MidAsk files there the data is stored every 5 minutes for each identifier from the list:

```sql
/****** Script for SelectTopNRows command from SSMS  ******/
SELECT TOP (1000) [FileID]
      ,[#RIC]
      ,[Alias Underlying RIC]
      ,[Domain]
      ,[Date-Time]
      ,[GMT Offset]
      ,[Type]
      ,[High Bid]
      ,[Low Bid]
      ,[Close Bid]
      ,[High Ask]
      ,[Low Ask]
      ,[Close Ask]
```

33 %

Results | Messages

	FileID	#RIC	Alias Underlying RIC	Domain	Date-Time	GMT Offset	Type	High Bid	Low Bid	Close Bid	High Ask	Low Ask	Close Ask
1	221	TFMBQc2	TFMBQM3	Market Price	2022-10-17T13:10:00.000000000Z	2	Intraday 5Min	151.17	151.165	151.17	152.425	151.945	152.215
2	221	TFMBQc2	TFMBQM3	Market Price	2022-10-17T13:15:00.000000000Z	2	Intraday 5Min	151.17	151.165	151.17	152.5	152.015	152.015
3	221	TFMBQc2	TFMBQM3	Market Price	2022-10-17T13:20:00.000000000Z	2	Intraday 5Min	151.17	148.805	150.49	152.215	151.41	151.625
4	221	TFMBQc2	TFMBQM3	Market Price	2022-10-17T13:25:00.000000000Z	2	Intraday 5Min	150.96	148.905	150.71	152.22	151.545	151.955
5	221	TFMBQc2	TFMBQM3	Market Price	2022-10-17T13:30:00.000000000Z	2	Intraday 5Min	151.535	149.145	150.055	153.49	150.495	150.99
6	221	TFMBQc2	TFMBQM3	Market Price	2022-10-17T13:35:00.000000000Z	2	Intraday 5Min	150.335	148.88	149.56	150.995	150.1	150.56
7	221	TFMBQc2	TFMBQM3	Market Price	2022-10-17T13:40:00.000000000Z	2	Intraday 5Min	149.605	148.805	149.41	150.79	150.23	150.24
8	221	TFMBQc2	TFMBQM3	Market Price	2022-10-17T13:45:00.000000000Z	2	Intraday 5Min	149.415	148.885	149.415	150.5	149.94	150.035
9	221	TFMBQc2	TFMBQM3	Market Price	2022-10-17T13:50:00.000000000Z	2	Intraday 5Min	150.165	148.17	149.37	150.69	150.015	150.4
10	221	TFMBQc2	TFMBQM3	Market Price	2022-10-17T13:55:00.000000000Z	2	Intraday 5Min	150	148.665	149.91	150.965	150.395	150.735
11	221	TFMBQc2	TFMBQM3	Market Price	2022-10-17T14:00:00.000000000Z	2	Intraday 5Min	150.665	148.56	150.51	152.775	150.4	151.15
12	221	TFMBQc2	TFMBQM3	Market Price	2022-10-17T14:05:00.000000000Z	2	Intraday 5Min	150.555	149.03	149.355	152.71	150.59	150.845
13	221	TFMBQc2	TFMBQM3	Market Price	2022-10-17T14:10:00.000000000Z	2	Intraday 5Min	150.59	149.355	149.815	152.96	150.67	151.735
14	221	TFMBQc2	TFMBQM3	Market Price	2022-10-17T14:15:00.000000000Z	2	Intraday 5Min	151.08	149.715	149.76	152.955	151.06	151.32
15	221	TFMBQc2	TFMBQM3	Market Price	2022-10-17T14:20:00.000000000Z	2	Intraday 5Min	149.76	148.87	149.08	151.385	149.99	149.995
16	221	TFMBQc2	TFMBQM3	Market Price	2022-10-17T14:25:00.000000000Z	2	Intraday 5Min	149.955	148.715	149.175	151.405	149.99	150.78
17	221	TFMBQc2	TFMBQM3	Market Price	2022-10-17T14:30:00.000000000Z	2	Intraday 5Min	149.505	148.925	149.055	151.295	150.335	150.335

Query executed successfully. | dw-svr01.database.windows.n... | svc_devadmins (72) | EXTERNAL_PARTIES | 00:00:00 | 1,000 rows

For LocationalSpread files the data is stored per day

```sql
/****** Script for SelectTopNRows command from SSMS  ******/
SELECT TOP (1000) [FileID]
      ,[RIC]
      ,[Trade Date]
      ,[Universal Close Price]
  FROM [Refinitiv].[LocationalSpreadData]
```

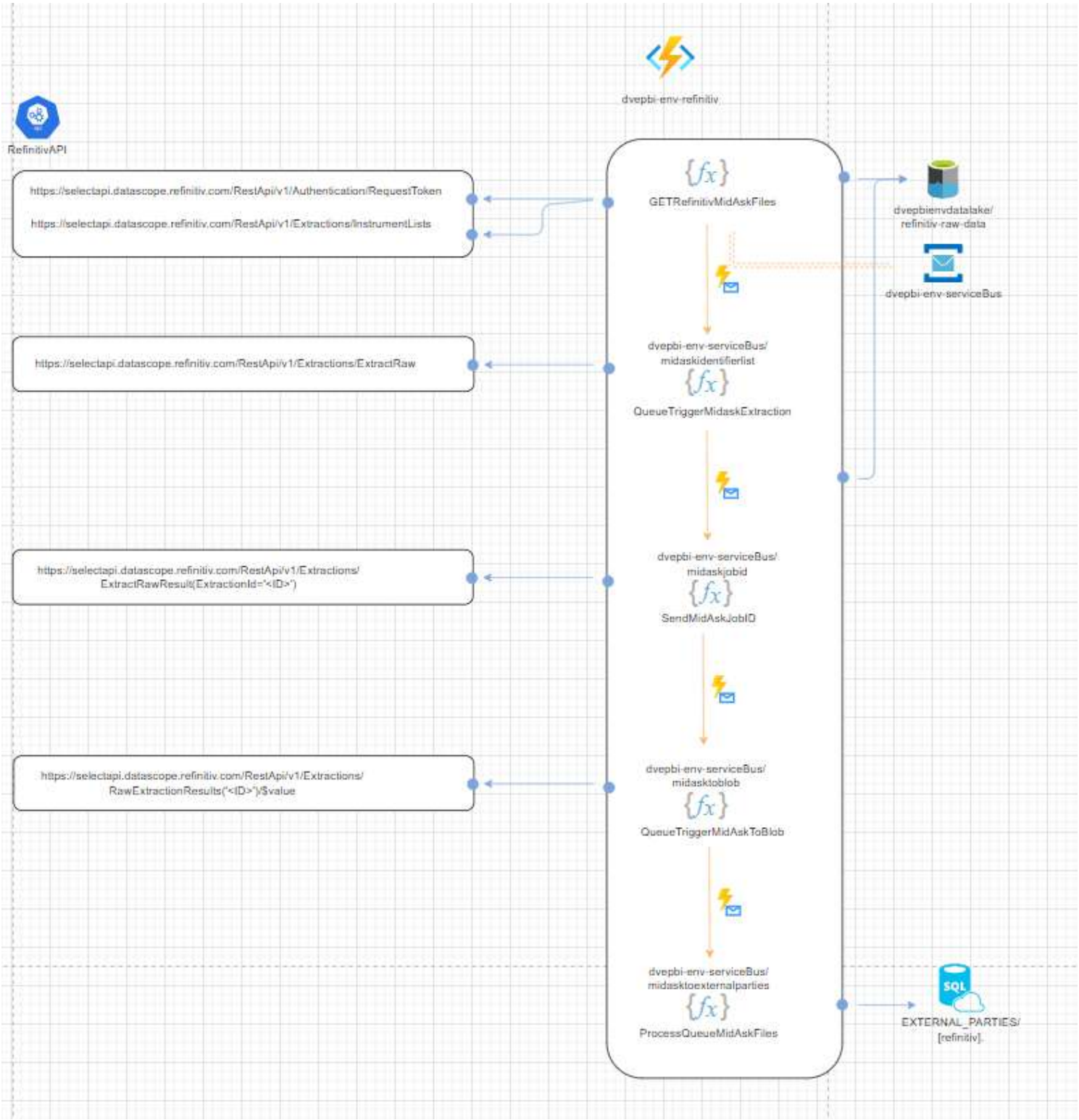
133 %

Results | Messages

	FileID	RIC	Trade Date	Universal Close Price
1	70	ZTPDA	2022/10/17	19
2	82	ZTPLDA	2022/10/19	67
3	108	PEGDA	2022/10/24	23.5
4	124	PPEGMc1	2022/10/26	88.7
5	141	PPEGMc1	2022/10/28	97.2
6	158	ZTPDA	2022/11/01	18.925
7	169	ZTPLDA	2022/11/03	70
8	185	ZTPLDA	2022/11/07	61.5
9	83	PTTFMc1	2022/10/19	112.32
10	200	TTFDA	2022/11/08	89.975
11	115	ZTPMc1	2022/10/25	91.32
12	209	ZTPLDA	2022/11/10	70
13	221	PTTFMc1	2022/11/11	97.924
14	93	PTTFMc1	2022/10/20	126.828
15	231	ZTPDA	2022/11/14	116.05
16	161	ZTPMc1	2022/11/02	122.37
17	177	ZTPDA	2022/11/04	44.8

For each of these two types we created a separate process.

MidAsk process:



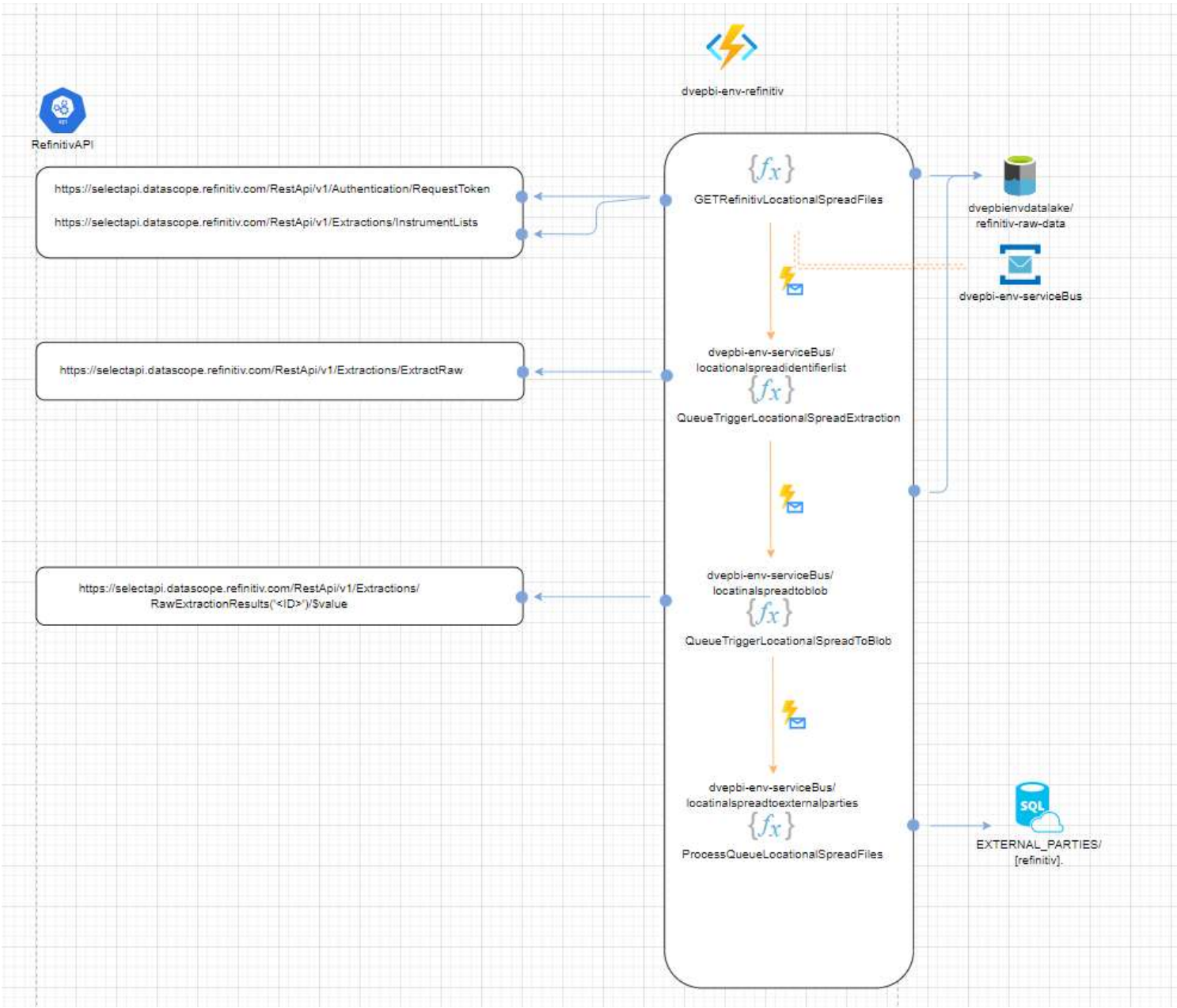
This process is using 5 Azure functions:

1. **GETRefinitivMidAskFiles**
   Is a timer trigger function which will run each day at 10:10DT from Monday to Friday. It's calling two APIs from Refinitiv one is for getting an active token and the other one for getting the list of the identifiers which are needed to extract the data. And sends each identifier to the QueueTriggerMidaskExtraction function using a queue message in ServiceBus.
2. **QueueTriggerMidaskExtraction**
   Is a SercviceBus trigger function which contains the API call which initiate the extraction of the data and

send the ID of the extraction to the SendMidAskJobID function using a queue message in ServiceBus.

3. **SendMidAskJobID**
   Is a SercviceBus trigger function which contains the API call which checks the status of the extraction, when the extraction is done it sends the ID to the QueueTriggerMidAskToBlob function using a queue message in ServiceBus.
4. **QueueTriggerMidAskToBlob**
   Is a SercviceBus trigger function which contains the API call which return the data extracted and save it to Blob Storage and send the name of the file to the ProcessQueueMidAskFiles function using a queue message in ServiceBus.
5. **ProcessQueueMidAskFiles**
   Is a SercviceBus trigger function function which load the data from Blob Storage in to the ExternalParties database.

LocationalSpread data:



This process is using 4 Azure functions:

1. **GETRefinitivLocationalSpreadFiles**
   Is a timer trigger function which will run each day at 10:00DT from Monday to Friday. It's calling two APIs from Refinitiv one is for getting an active token and the other one for getting the list of the identifiers which are needed to extract the data. And sends each identifier to the QueueTriggerLocationalSpreadExtraction function using a queue message in ServiceBus.
2. **QueueTriggerLocationalSpreadExtraction**
   Is a SercviceBus trigger function which contains the API call which initiate the extraction of the data and send the ID of the extraction to the QueueTriggerLocationalSpreadToBlob function using a queue message in ServiceBus.
3. **QueueTriggerLocationalSpreadToBlob**
   Is a SercviceBus trigger function which contains the API call which return the data extracted and save it to Blob Storage and send the name of the file to the ProcessQueueLocationalSpreadFiles function using a queue message in ServiceBus.
4. **ProcessQueueLocationalSpreadFiles**
   Is a SercviceBus trigger function which load the data from Blob Storage in to the ExternalParties database.